

Electrostatics

• Electric charge

↳ Electric charge is the intrinsic property of a matter due to which it produces and experiences electrical and magnetic effects.

• Properties of Electric charge

↳ The charge is a scalar quantity, so, it doesn't obey vector law of algebra.

↳ like charges repel each other but unlike charges attract each other.

↳ The charge follows conservation principle i.e. the total charge of an isolated system remains constant. This means charge can be neither created nor destroyed but transferred from one body to another.

↳ The charge is always in the form of an integral multiple of electronic charge and never its fraction. This is known as quantization of charge.

i.e. $q = \pm ne$ where, $n = 1, 2, 3, \dots$ is integer and $e = 1.6 \times 10^{-19} \text{ C}$ is the charge of electron or proton.

- Methods of charging a Body
→ An object can be electrified by different methods as given below:

① By Friction

- When a body is rubbed with another suitable pair give and take of electrons occurs and the bodies become charged.

The body that loses electrons acquires a positive charge and another that receives electrons possesses a negative charge. Example: glass, fur etc.

② By Conduction.

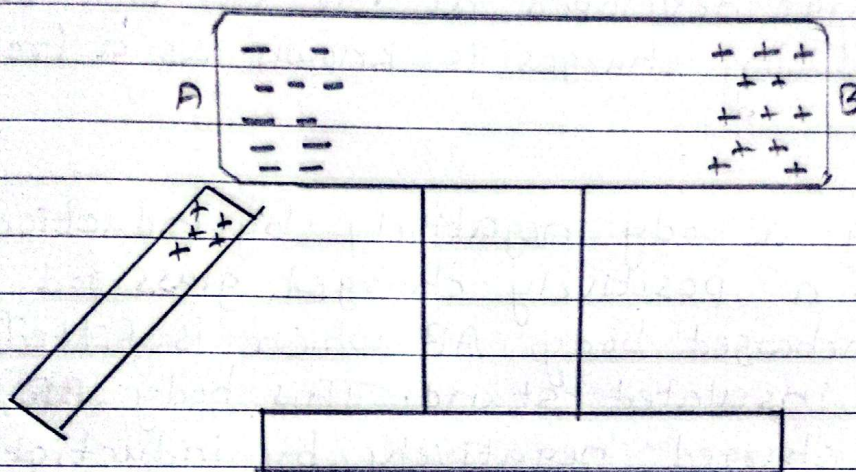
- When a neutral body is brought into contact with a charged body some charges flow into it and becomes charged. The charge acquired by the neutral body is similar to that of the charged body.

③ By induction

- The process of development of electric charge at the two ends of an isolated conductor, when a charged body is brought near one end of the conductor is called electrostatic induction.

• Electrostatic induction

→ The process of temporary electrification of an initially uncharged conducting body by bringing a charged body close to it but not making actual contact is known as electrostatic induction.



• Inducing charge

→ The charge present in a charged body that induces charges in another uncharged body is called inducing charge.

• Induced charge

→ The charge developed in an initially uncharged conducting body by bringing a charged body close to it but not making actual contact is known as induced charge.

These are of two types

- ① Bound charge
- ② Free charge

- Bound charge

→ The charges developed at the end closer to the inducing charge are known as a bound charge.

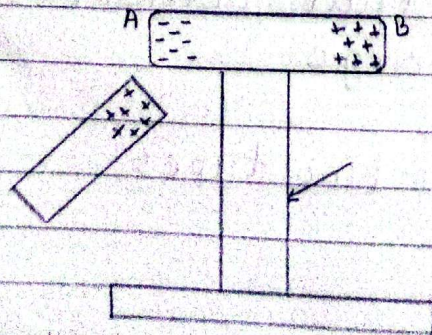
- Free charge

→ The charge developed at the far end to the inducing charges is known as a free charge.

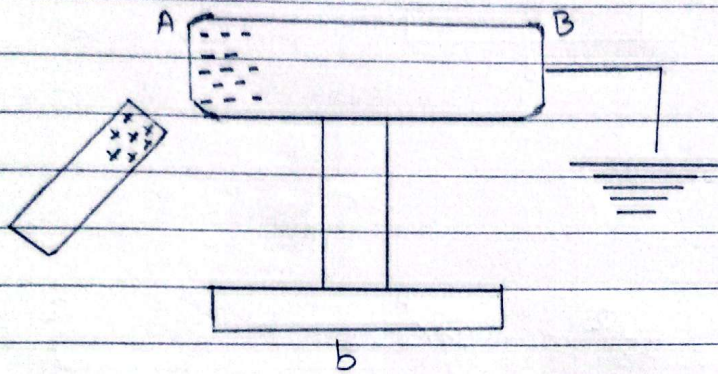
- Charging a body negatively by induction

- Consider a positively charged glass rod and uncharged body AB which is fitted on the insulated stand. This body AB can be charged negatively by induction in the following ways:

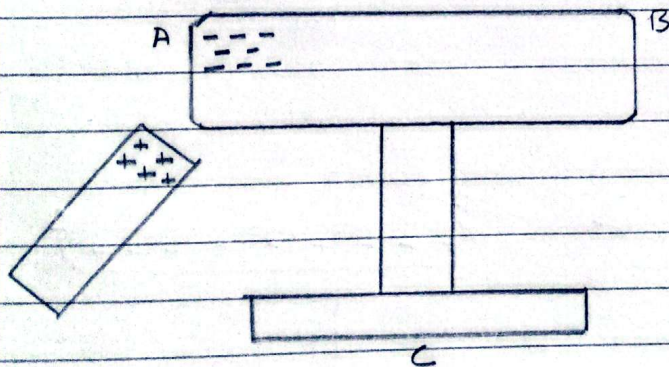
a) When a positively charged glass rod is brought near a body AB, the end A of the body acquires bound negative charge and the far end B acquires free positive charge due to induction as in fig (a)



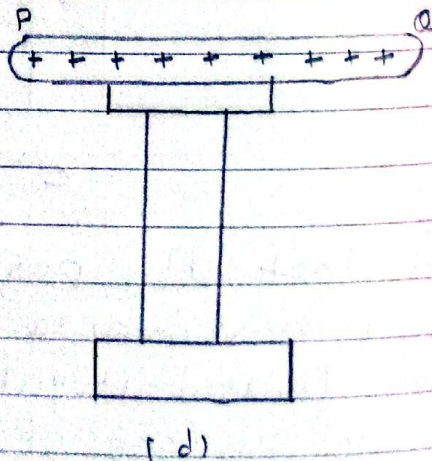
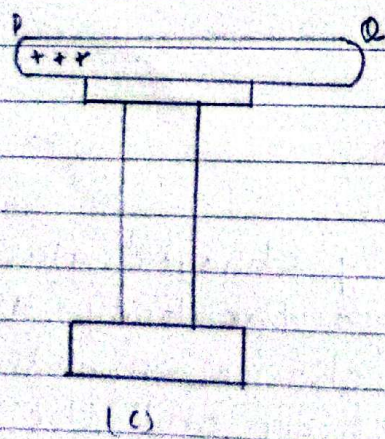
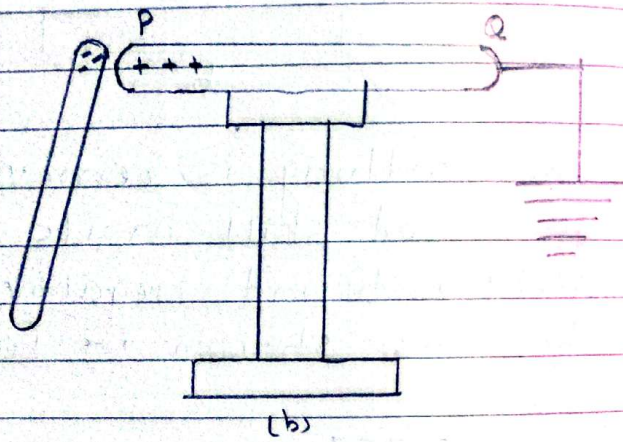
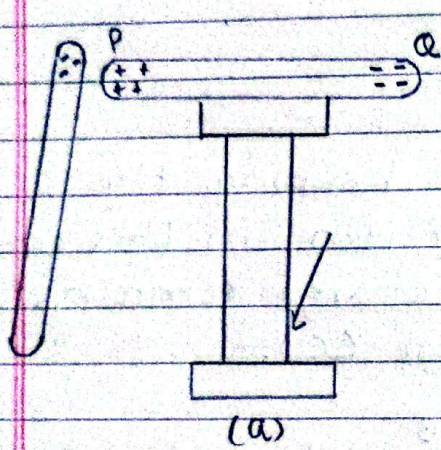
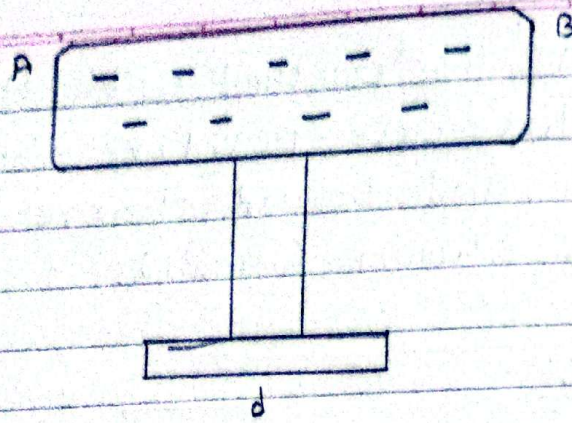
- b) The body AB is earthed with the help of wire. In this case positive charge gets neutralized and bound charge remains as it is as shown fig (b).



- c) The earthing is removed keeping the glass rod still in its position, in this condition bound negative charge remains there as shown in figure (c).



- d) At last, the positively charged glass rod is removed away from the body AB. The bound negative charge spreads over whole surface of the body and the body AB becomes negatively charged body by induction as shown in figure (d).



• Conductors and Insulators

↳ The substances which can conduct electricity through them easily are called conductor. For example, silver, gold, copper, human body, earth etc. Such substances have sufficient free electrons and the charge are not localized but uniformly distributed over their surfaces.

The substances which do not conduct electricity through them are called insulators. For example wood, plastics, diamond, glass, paper etc. In such substances, there are not free electrons. Consequently, the electric charges are not uniformly distributed over their surfaces but localize.

• Surface Density of charge

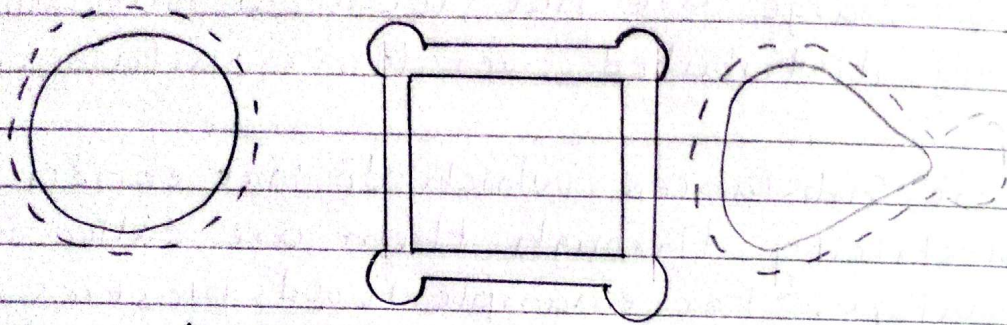
↳ The Surface density of charge is defined as the amount of charge per unit area of a charged surface. It is denoted by σ (sigma).

If Q amount of charge is distributed uniformly over the surface area A of a conductor, then the surface density of a charge is given by

$$\sigma = \frac{\text{charge } (Q)}{\text{area } (A)}$$

i.e. $\sigma = \frac{Q}{A}$

In SI units the units of surface density of charge is ~~cm~~ coulomb meter⁻² (Cm⁻²). The surface charge density depends upon the shape of the conductor.



• Action of Points

→ The process of losing of charge from the sharp end of the conductor is called action of points.

Qn. Why are sharp edges or points avoided in electrical machine?

→ For uniform shaped body, distribution of charge is uniform but for other shaped body the distribution of charge is inversely proportional to the radii of curvature of the surfaces. So surface charge density is maximum at the pointed ends of a conductor. When air molecules come in contact to the pointed ends of a charged conductor, they get similar charged and are repelled. So large amount

of charge is lost due to action of points. Thus, to avoid the leakage of charge from the electrical machines, sharp edges or points are strictly avoided.

• Coulomb's Law

↳ It states that "the force of attraction or repulsion between two point charges is directly proportional to the product of charges and inversely proportional to the square of the distance between them."

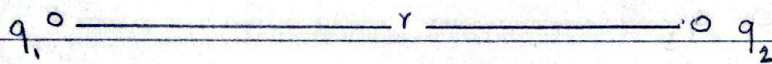


fig: two charges separated by distance r

Mathematically,

$$F \propto q_1 q_2 \rightarrow (i)$$

$$F \propto \frac{1}{r^2} \rightarrow (ii)$$

Combining equation (i) & (ii), we get

$$F \propto \frac{q_1 q_2}{r^2}$$

$$F = k \frac{q_1 q_2}{r^2}$$

Here, k is the proportionality constant

whose value depends upon the medium in which charges are present and the system of a unit chosen.

For S.I. system and charges present in the vacuum or air.

$$K = \frac{1}{4\pi\epsilon_0} \quad \text{Here, } \epsilon_0 \text{ is permittivity of vacuum or free space.}$$

$$\therefore F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

In CGS, $K=1$

$$\therefore F = \frac{q_1 q_2}{r^2} \quad \text{H.}$$

• Permittivity (ϵ)

↳ It is the property of every material, which measures the opposition offered against the formation of an electric field. Represented by the Greek alphabet ϵ . ~~It~~ It tells the number of charges required to generate one unit of electric flux in the given medium. Its unit is Farad / meter.

The vacuum characterizes the least possible value of permittivity. This is com-

only referred to as the permittivity of free space or electric constant. Denoted by ϵ_0 and has the value 8.85×10^{-12} Farad / meter.

- Relative Permittivity or dielectric constant (ϵ_r)

↳ The dielectric constant or relative permittivity of a medium can be defined as the ratio of the permittivity of a medium (ϵ) to the permittivity of vacuum or free space (ϵ_0)

$$\epsilon_r = \frac{\epsilon}{\epsilon_0}$$

In terms of force between charges, the dielectric constant of a medium can be defined as the ratio of force between two charges at certain distance in vacuum and the force between the same charges placed at the same distance in such medium